

Absolute Normalization of the Compact $c = 1$ String Ingredient by Ingredient in Xi's Worldsheet Conventions

Scratch derivation from the action, CFT data, and sewing unitarity

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Abstract

We derive the genus-one and genus-two vacuum integrands of the compact $c = 1$ bosonic string using the logic of the string notes. Every ingredient is computed separately: the compact zero mode and lattice action, oscillator determinants, Liouville spectral measure, ghost determinant, moduli form, loop-momentum Gaussian, Mumford-form residue, topology sewing constant, and global stack weight. The matrix model is not used as a normalization input. It is introduced only after the worldsheet answer has been fixed, as an external comparison.

The main bookkeeping distinction is between the universal BRST normalization $K_{S^2}^{\text{crit}} = 8\pi/\alpha'$ used in the critical-string measure and the reduced physical $c = 1$ sphere normalization $\hat{K}_{S^2}^{c=1} = 2\pi\sqrt{\alpha'} \tilde{K}_{S^2} = 4\pi$. Sewing then gives

$$\frac{C_g^{c=1}}{C_g^{\text{crit}}} = \left(\frac{2}{\alpha'}\right)^{g-1}.$$

The torus vacuum is unchanged, while the genus-two local kernel is multiplied by $2/\alpha'$. At $\alpha' = 1$ the positive-real, g_s^2 -stripped coefficient is $2/\pi$.

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1 Logic contract: what is input and what is derived

The purpose of this note is to prevent an overall constant from entering through an implicit convention. Every numerical factor must belong to one of the following categories.

Minimal input 1 (Worldsheet theory). *The matter theory is a timelike free boson X^0 times $b = 1$ Liouville theory, together with the standard (b, c) ghosts and a constant dilaton Φ_0 . We set $g_s = e^{\Phi_0}$.*

Minimal input 2 (Ghost normalization). *On the sphere,*

$$\left\langle \prod_{i=1}^3 c(z_i) \tilde{c}(\bar{z}_i) \right\rangle_{S^2} = |z_{12} z_{13} z_{23}|^2. \quad (1.1)$$

This is Xi's equation (4.32).

Minimal input 3 (Liouville states). *The reflection-symmetric Liouville primary V_P , $P \geq 0$, is normalized by*

$$\langle V_P(1) V_{P'}(0) \rangle = \pi \delta(P - P'), \quad \mathbf{1}_{\mathcal{H}_L} = \int_0^\infty \frac{dP}{\pi} |P\rangle \langle P|, \quad (1.2)$$

and

$$\langle V_{P_1} V_{P_2} V_{P_3} \rangle = C(P_1, P_2, P_3). \quad (1.3)$$

These are Xi's equations (I.39)–(I.42).

Minimal input 4 (Physical string states). *The one-string in/out states and their vertices are*

$${}_{\text{in/out}} \langle \omega | \omega' \rangle_{\text{in/out}} = \omega \delta(\omega - \omega'), \quad (1.4)$$

$$\mathcal{V}_\omega^\pm = g_s c \tilde{c} \exp\left(\pm \frac{i\omega X^0}{\sqrt{\alpha'}}\right) V_{P=\omega/2}. \quad (1.5)$$

These are Xi's equations (4.111)–(4.112).

Minimal input 5 (Plumbing coordinate and differential forms). *Local coordinates are sewn literally by*

$$uv = q, \quad q = e^{2\pi i \tau}. \quad (1.6)$$

The cylinder has circumference 2π . Differential forms are ordered as in the string notes; in particular

$$i dz \wedge d\bar{z} = 2 d(\Re z) d(\Im z). \quad (1.7)$$

Minimal input 6 (Unitarity). *With $S = 1 + iT$, impose $SS^\dagger = 1$. A one-particle pole must factorize with the inverse of the physical two-point metric. On the worldsheet this pole is produced by the region $q \rightarrow 0$ of moduli space.*

No matrix-model free energy, no numerical Monte Carlo result, and no guessed integer or power of 2 will be used below.

2 Worldsheet conventions

2.1 Action and central charge

In Lorentzian target signature, the matter action is

$$S_{X^0} = -\frac{1}{4\pi\alpha'} \int_{\Sigma} d^2\sigma \sqrt{g} g^{ab} \partial_a X^0 \partial_b X^0, \quad (2.1)$$

$$S_L[X^1] = \frac{1}{4\pi\alpha'} \int_{\Sigma} d^2\sigma \sqrt{g} \left[g^{ab} \partial_a X^1 \partial_b X^1 + \sqrt{\alpha'} Q X^1 R(g) + 4\pi\mu_L e^{2bX^1/\sqrt{\alpha'}} \right]. \quad (2.2)$$

Here $Q = b + b^{-1}$. At $b = 1$, $Q = 2$ and

$$c_{X^0} + c_L + c_{bc} = 1 + 25 - 26 = 0. \quad (2.3)$$

The constant dilaton contributes

$$S_{\Phi_0} = \Phi_0 \chi(\Sigma), \quad e^{-S_{\Phi_0}} = g_s^{2g-2}. \quad (2.4)$$

Since each physical vertex in (1.5) contains one g_s , a genus- g , n -point amplitude is of order g_s^{2g-2+n} .

For the thermal vacuum amplitude we Wick rotate X^0 to a Euclidean compact boson X with

$$X \sim X + 2\pi R_{\text{phys}}, \quad r := \frac{R_{\text{phys}}}{\sqrt{\alpha'}}. \quad (2.5)$$

The worldsheet torus coordinate has periods 2π and $2\pi\tau$.

The action also fixes the conformal weights used in the sewing calculation. With the exponential in (1.5),

$$h_{X^0} = \bar{h}_{X^0} = -\frac{\omega^2}{4}, \quad h_L = \bar{h}_L = \frac{Q^2}{4} + P^2 = 1 + P^2 \quad (b = 1). \quad (2.6)$$

Thus the product is a $(1, 1)$ primary precisely when $P = \omega/2$. Off this physical locus the quantity that appears in a plumbing propagator is

$$a(P, \omega) = P^2 - \frac{\omega^2}{4}. \quad (2.7)$$

2.2 The special $b = 1$ three-point identity

The DOZZ coefficient in the normalization (1.2) obeys

$$C(P_1, P_2, P_1 + P_2) = 8P_1 P_2 (P_1 + P_2). \quad (2.8)$$

On the physical kinematic locus $P_i = \omega_i/2$ and $\omega = \omega_1 + \omega_2$, this becomes

$$C\left(\frac{\omega}{2}, \frac{\omega_1}{2}, \frac{\omega_2}{2}\right) = \omega \omega_1 \omega_2. \quad (2.9)$$

This identity fixes the local three-string interaction once the sphere zero mode and the string coupling are normalized.

3 Unitarity and the sphere constants

3.1 Critical BRST state metric

Let $N_{g,n}$ multiply Xi's top differential form on $\mathcal{M}_{g,n}$. Plumbing two surfaces of types (g_1, n_1) and (g_2, n_2) and matching the pole to the one-particle unitarity sum gives

$$N_{g_1, n_1} N_{g_2, n_2} = -i N_{g, n} \frac{8\pi}{\alpha' K_{S^2}^{\text{crit}}}, \quad g = g_1 + g_2, \quad n = n_1 + n_2 - 2. \quad (3.1)$$

Pinching one handle gives

$$N_{g-1, n+2} = -i N_{g, n} \frac{8\pi}{\alpha' K_{S^2}^{\text{crit}}}. \quad (3.2)$$

These are Xi's equations (4.69)–(4.70). The simultaneous solution adopted in the notes is

$$\boxed{N_{g, n} = i^{3g-3+n}, \quad K_{S^2}^{\text{crit}} = \frac{8\pi}{\alpha'}.} \quad (3.3)$$

To see that the numerical value of $K_{S^2}^{\text{crit}}$ is fixed rather than assumed, substitute the phase ansatz into the separating recurrence. Since $g = g_1 + g_2$ and $n = n_1 + n_2 - 2$,

$$\frac{N_{g_1, n_1} N_{g_2, n_2}}{N_{g, n}} = i^{-1} = -i. \quad (3.4)$$

Comparison with (3.1) therefore requires $8\pi/(\alpha' K_{S^2}^{\text{crit}}) = 1$, which is exactly the second result in (3.3). The nonseparating recurrence gives the same condition. In particular,

$$N_{1,0} = 1, \quad N_{2,0} = -i. \quad (3.5)$$

3.2 Physical $c = 1$ timelike sphere normalization

In the $c = 1$ scattering convention, write the timelike zero-mode correlator as

$$\left\langle e^{-ik^0 X^0} \right\rangle_{S^2, X^0} = 2\pi i \delta(k^0) g_s^{-2} \tilde{K}_{S^2}. \quad (3.6)$$

Here is the pole calculation behind the normalization. The literal fixture $uv = q$ gives, for the radial part of a closed-string propagator,

$$\int_{|q|<1} \frac{d^2 q}{2\pi |q|^2} |q|^{2a} = \frac{1}{2a}, \quad a = P^2 - \frac{\omega^2}{4}. \quad (3.7)$$

The Liouville completeness measure then has the on-shell discontinuity

$$\text{Im} \int_0^\infty \frac{dP}{\pi} \frac{1}{P^2 - \omega^2/4 - i0} = \int_0^\infty dP \delta\left(P^2 - \frac{\omega^2}{4}\right) = \frac{1}{\omega}. \quad (3.8)$$

This is exactly the inverse metric required by $\langle \omega | \omega' \rangle = \omega \delta(\omega - \omega')$. In Xi's closed-string propagator convention the remaining zero-mode residue is $\sqrt{\alpha'} \tilde{K}_{S^2}/2$; the factor $1/2$ is the radial residue in (3.7). Requiring unit coefficient gives

$$\frac{\sqrt{\alpha'} \tilde{K}_{S^2}}{2} = 1, \quad (3.9)$$

and hence

$$\boxed{\tilde{K}_{S^2} = \frac{2}{\sqrt{\alpha'}}}. \quad (3.10)$$

This is Xi's equation (4.114). The momentum in (3.6) is physical,

$$k^0 = \frac{\omega}{\sqrt{\alpha'}}, \quad \delta(k^0) = \sqrt{\alpha'} \delta(\omega). \quad (3.11)$$

Consequently the coefficient of the dimensionless energy delta function is

$$\boxed{\hat{K}_{S^2}^{c=1} := 2\pi\sqrt{\alpha'} \tilde{K}_{S^2} = 4\pi}. \quad (3.12)$$

Convention warning 1. $K_{S^2}^{\text{crit}}$ and $\tilde{K}_{S^2}$ are not two names for the same quantity. The first normalizes the universal BRST state metric in the critical-string derivation. The second is the timelike zero-mode constant in the physical $c = 1$ scattering normalization. After converting to the dimensionless energy delta function, the $c = 1$ topology metric is $\hat{K}$.

3.3 Tree-level $1 \rightarrow 2$ check

Three vertices give g_s^3 and the sphere dilaton factor gives g_s^{-2} . Using (3.12) and (2.9), the coefficient of $\delta(\omega - \omega_1 - \omega_2)$ is

$$\begin{aligned} \mathcal{A}_0(\omega_1, \omega_2) &= i g_s \hat{K}_{S^2}^{c=1} C\left(\frac{\omega}{2}, \frac{\omega_1}{2}, \frac{\omega_2}{2}\right) \\ &= \boxed{4\pi i g_s \omega \omega_1 \omega_2}, \quad \omega = \omega_1 + \omega_2. \end{aligned} \quad (3.13)$$

This reproduces Xi's equation (4.121) without using the matrix model.

4 Topology constants from sewing

First strip the single energy-conservation delta function and the local CFT density from every connected amplitude. Denote the remaining topology constant by C_g . With this reduced-amplitude convention, the tree result (3.13) says directly that

$$C_0^{c=1} = \hat{K}_{S^2}^{c=1} = 4\pi. \quad (4.1)$$

Thus the Fourier 2π and the Jacobian in (3.11) are part of the reduced sphere coefficient, whereas the polynomial $C(\omega/2, \omega_1/2, \omega_2/2)$ remains in the local CFT density.

Sewing a genus- g_1 and a genus- g_2 surface through one inverse two-point metric gives

$$C_{g_1+g_2} K_{S^2} = C_{g_1} C_{g_2}. \quad (4.2)$$

The canonical torus trace fixes $C_1 = 1$. Taking $g_2 = 1$ recursively, or solving (4.2) directly, gives

$$C_g = (K_{S^2})^{1-g}, \quad C_0 = K_{S^2}. \quad (4.3)$$

Therefore replacing the critical sphere metric by the physical $c = 1$ metric changes a genus- g vacuum density by

$$\Lambda_g := \frac{C_g^{c=1}}{C_g^{\text{crit}}} = \left(\frac{K_{S^2}^{\text{crit}}}{\widehat{K}_{S^2}^{c=1}} \right)^{g-1} = \left(\frac{2}{\alpha'} \right)^{g-1}. \quad (4.4)$$

Two immediate consequences are

$$\Lambda_1 = 1, \quad \Lambda_2 = \frac{2}{\alpha'}. \quad (4.5)$$

This is the precise sense in which a genus-one vacuum computation cannot see the sphere-normalization mismatch: the torus has zero Euler characteristic. A constant rescaling in the g_s - μ dictionary may shift a regulated $\log \mu$ by an additive constant, but it cannot multiply the universal coefficient of $\log \mu$.

5 Genus one: derivation of every ingredient

5.1 Geometry and classical compact-boson sectors

Let $\sigma^1, \sigma^2 \in [0, 2\pi)$ and $z = \sigma^1 + \tau\sigma^2$. In these coordinates the flat torus metric is

$$g_{ab} = \begin{pmatrix} 1 & \tau_1 \\ \tau_1 & |\tau|^2 \end{pmatrix}, \quad \sqrt{g} = \tau_2, \quad g^{ab} = \frac{1}{\tau_2^2} \begin{pmatrix} |\tau|^2 & -\tau_1 \\ -\tau_1 & 1 \end{pmatrix}. \quad (5.1)$$

A map to the target circle can wind independently around the two cycles:

$$X_{m,n} = x_0 + R_{\text{phys}}(m\sigma^1 + n\sigma^2), \quad m, n \in \mathbb{Z}. \quad (5.2)$$

Substitution into the Euclidean version of (2.1) gives

$$\begin{aligned} S_{m,n} &= \frac{(2\pi)^2}{4\pi\alpha'} \tau_2 R_{\text{phys}}^2(m, n) g^{-1} \binom{m}{n} \\ &= \frac{\pi R_{\text{phys}}^2}{\alpha' \tau_2} (|\tau|^2 m^2 - 2\tau_1 mn + n^2) = \frac{\pi R_{\text{phys}}^2}{\alpha' \tau_2} |m\tau - n|^2. \end{aligned} \quad (5.3)$$

This calculation fixes both the π and the placement of τ_2 in the lattice exponential.

5.2 Compact zero mode and oscillator determinant

The nonzero modes of one real scalar give

$$Z_X^{\text{osc}}(\tau) = \tau_2^{-1/2} |q|^{-1/12} \prod_{k=1}^{\infty} |1 - q^k|^{-2} = \frac{1}{\sqrt{\tau_2} |\eta(\tau)|^2}, \quad q = e^{2\pi i \tau}. \quad (5.4)$$

The zero-mode measure compatible with Xi's noncompact momentum measure $dk/(2\pi)$ is

$$\int_0^{2\pi R_{\text{phys}}} \frac{dx_0}{2\pi\sqrt{\alpha'}} = \frac{R_{\text{phys}}}{\sqrt{\alpha'}}. \quad (5.5)$$

This is not an additional convention: it is forced by the canonical trace. Indeed, with $p_{L,R} = n/R_{\text{phys}} \pm wR_{\text{phys}}/\alpha'$, Poisson resummation uses

$$\sum_{n \in \mathbb{Z}} e^{-\pi\tau_2\alpha'n^2/R_{\text{phys}}^2 + 2\pi i\tau_1 nw} = \frac{R_{\text{phys}}}{\sqrt{\alpha'\tau_2}} \sum_{\tilde{n} \in \mathbb{Z}} e^{-\pi R_{\text{phys}}^2(\tilde{n} - w\tau_1)^2/(\alpha'\tau_2)}, \quad (5.6)$$

which produces exactly the same zero-mode coefficient. Combining (5.3)–(5.5),

$$\boxed{Z_X(R_{\text{phys}}; \tau) = \frac{R_{\text{phys}}}{\sqrt{\alpha'\tau_2}|\eta(\tau)|^2} \sum_{m,n \in \mathbb{Z}} \exp\left[-\frac{\pi R_{\text{phys}}^2}{\alpha'\tau_2}|m\tau - n|^2\right]}. \quad (5.7)$$

5.3 Liouville spectral trace

The normalized Liouville identity resolution (1.2) restricts the reflection-symmetric spectrum to $P \geq 0$. In Xi's dimensionful momentum, $P_{\text{note}} = 2P/\sqrt{\alpha'}$, the character is $e^{-\pi\alpha'\tau_2 P_{\text{note}}^2/|\eta|^2}$. Therefore

$$\begin{aligned} \frac{Z_L(\tau)}{V_\phi} &= \int_0^\infty \frac{dP_{\text{note}}}{\pi} \frac{e^{-\pi\alpha'\tau_2 P_{\text{note}}^2}}{|\eta(\tau)|^2} \\ &= \frac{1}{\pi|\eta|^2} \frac{\sqrt{\pi}}{2\sqrt{\pi\alpha'\tau_2}} = \boxed{\frac{1}{2\pi\sqrt{\alpha'\tau_2}|\eta(\tau)|^2}}. \end{aligned} \quad (5.8)$$

The factor $1/2$ is thus the half-line Gaussian, not an automorphism factor.

5.4 Ghosts, modulus, conformal Killing volume, and stack weight

The chiral nonzero-mode trace of the weight- $(2, -1)$ ghost system is $\eta(\tau)^2$; the left-right product is $|\eta|^4$. The b zero modes are saturated by the Beltrami insertions for τ and $\bar{\tau}$. With the sphere ghost normalization (1.1), the resulting unmarked torus formula is Xi's equation (4.90),

$$\mathcal{A}_1 = \frac{1}{2} \int_{\mathcal{F}_1} \underbrace{\frac{i d\tau \wedge d\bar{\tau}}{2\tau_2}}_{\text{modulus and translation CKV}} \underbrace{|\eta(\tau)|^4}_{bc \text{ nonzero modes}} Z_m(\tau, \bar{\tau}). \quad (5.9)$$

Here $N_{1,0} = 1$ and $g_s^{2g-2} = 1$. The prefactor $1/2$ is the generic $z \mapsto -z$ stabilizer of an unmarked torus when $\mathcal{F}_1$ is a coarse $PSL(2, \mathbb{Z})$ fundamental domain. It is distinct from the translation CKV factor $1/(2\tau_2)$. Finally,

$$i d\tau \wedge d\bar{\tau} = 2 d^2\tau \implies \frac{i d\tau \wedge d\bar{\tau}}{2\tau_2} = \frac{d^2\tau}{\tau_2}. \quad (5.10)$$

5.5 Multiplication: the complete genus-one integrand

Insert $Z_m = Z_X Z_L$ into (5.9). Every factor can be seen before cancellation:

$$\frac{d\mathcal{A}_1}{V_\phi} = \underbrace{\frac{1}{2}}_{\text{stack modulus/CKV}} \underbrace{\frac{d^2\tau}{\tau_2}}_{\text{modulus/CKV}} \underbrace{|\eta|^4}_{bc} \underbrace{\frac{R_{\text{phys}}}{\sqrt{\alpha'\tau_2}|\eta|^2}}_X \underbrace{\Theta_1}_{\text{ghost}} \underbrace{\frac{1}{2\pi\sqrt{\alpha'\tau_2}|\eta|^2}}_L, \quad (5.11)$$

$$\Theta_1 := \sum_{m,n \in \mathbb{Z}} \exp \left[-\frac{\pi R_{\text{phys}}^2}{\alpha'\tau_2} |m\tau - n|^2 \right]. \quad (5.12)$$

Thus $|\eta|^{4-2-2} = 1$, the two factors $\tau_2^{-1/2}$ give τ_2^{-1} , and the moduli form supplies the second τ_2^{-1} . The answer is

$$\boxed{\frac{1}{V_\phi} \frac{d\mathcal{A}_1}{d^2\tau} = \frac{R_{\text{phys}}}{4\pi\alpha'\tau_2^2} \sum_{m,n \in \mathbb{Z}} \exp \left[-\frac{\pi R_{\text{phys}}^2}{\alpha'\tau_2} |m\tau - n|^2 \right]}. \quad (5.13)$$

This is the genus-one integrand used by the direct τ -plane computation. There is no sphere-topology correction because $\Lambda_1 = 1$.

5.6 Unfolding check and the universal logarithm

The zero orbit uses $\int_{\mathcal{F}_1} d^2\tau/\tau_2^2 = \pi/3$ and contributes $R_{\text{phys}}/(12\alpha')$. Every nonzero pair can be written as a primitive pair times $k \geq 1$; modular images unfold $\mathcal{F}_1$ to the strip. Hence

$$\begin{aligned} \left. \frac{\mathcal{A}_1}{V_\phi} \right|_{\text{nonzero}} &= \frac{R_{\text{phys}}}{4\pi\alpha'} 2 \sum_{k=1}^{\infty} \int_{-1/2}^{1/2} d\tau_1 \int_0^{\infty} \frac{d\tau_2}{\tau_2^2} e^{-\pi R_{\text{phys}}^2 k^2 / (\alpha'\tau_2)} \\ &= \frac{R_{\text{phys}}}{4\pi\alpha'} 2 \sum_{k=1}^{\infty} \frac{\alpha'}{\pi R_{\text{phys}}^2 k^2} = \frac{1}{12R_{\text{phys}}}. \end{aligned} \quad (5.14)$$

Therefore

$$\boxed{\frac{\mathcal{A}_1}{V_\phi} = \frac{1}{12} \left(\frac{R_{\text{phys}}}{\alpha'} + \frac{1}{R_{\text{phys}}} \right)}. \quad (5.15)$$

The Liouville potential depends on its zero mode through $\mu_L \exp(2bX^1/\sqrt{\alpha'})$. Shifting the zero mode to absorb μ_L gives

$$V_\phi^{\text{ren}} = -\frac{\sqrt{\alpha'}}{2b} \log \mu_L + C_{\text{cutoff}}. \quad (5.16)$$

At $b = 1$ and $\alpha' = 1$, where $r = R_{\text{phys}}$,

$$\mathcal{A}_1|_{\log \mu_L} = -\frac{1}{24} \left(r + \frac{1}{r} \right) \log \mu_L. \quad (5.17)$$

6 Genus two: derivation of every ingredient

6.1 Period-matrix coordinates and the ghost top form

Let

$$\Omega = X + iY = \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{12} & \Omega_{22} \end{pmatrix}, \quad Y \succ 0, \quad (6.1)$$

and order the three holomorphic differentials on moduli space as

$$d^3\Omega := d\Omega_{11} \wedge d\Omega_{22} \wedge d\Omega_{12}. \quad (6.2)$$

There are three b -ghost zero modes because $\dim_{\mathbb{C}} \mathcal{M}_2 = 3$. In the basis $\omega_1^2, \omega_2^2, \omega_1\omega_2$ of holomorphic quadratic differentials, contraction with the three Beltrami differentials converts the ghost zero-mode determinant into $d^3\Omega$. This is the content of Xi's equations (4.100)–(4.103); it explains why no separate insertion-point Jacobian remains in the vacuum measure.

For one complex coordinate $z = x + iy$, $dz \wedge d\bar{z} = -2i dx \wedge dy$. Reordering the three barred forms next to their unbarred partners takes three transpositions. Therefore

$$d^3\Omega \wedge d^3\bar{\Omega} = (-1)^3 (-2i)^3 d^3X d^3Y = -8i d^3X d^3Y. \quad (6.3)$$

The factor 8 is solely a complex-to-real form Jacobian.

6.2 Critical seed and the 26 loop-momentum Gaussians

Before doing the harmonic-momentum integrals, Xi's equation (4.105) is

$$f(\Sigma) Z_X^{26}(\Sigma) = V_X \frac{g_s^2 \alpha'}{8\pi} \int \prod_{I=1}^2 \frac{d^{26} k_I}{(2\pi)^{26}} e^{-\pi \alpha' k_I \cdot k_J Y_{IJ}} |\Phi(\Omega)|^2. \quad (6.4)$$

For one target component, diagonalizing the positive matrix Y gives

$$\int \frac{d^{26} k}{(2\pi)^{26}} e^{-\pi \alpha' k^T Y k} = \frac{1}{(2\pi)^2} \frac{1}{\sqrt{\det(\alpha' Y)}} = \frac{1}{4\pi^2 \alpha' \sqrt{\det Y}}. \quad (6.5)$$

There are 26 components, hence $(4\pi^2 \alpha')^{-26} (\det Y)^{-13}$. Rotating the two timelike loop momenta to Euclidean contours gives $i^2 = -1$. The complex six-form is therefore

$$\Omega_6^{\text{crit}} = -V_X \frac{g_s^2 \alpha'}{8\pi} \frac{|\Phi(\Omega) d^3\Omega|^2}{(4\pi^2 \alpha')^{26} (\det Y)^{13}}. \quad (6.6)$$

Convention warning 2. *The factor $g_s^2 \alpha' / (8\pi)$ displayed in Xi's equation (4.105) must remain after the Gaussian. It is absent from the printed equation (4.106), but the preceding equation and the gauge-fixing recurrence require it.*

6.3 Oscillators, the Igusa form, and the factor 2^{24}

The combined chiral oscillator function of the ghosts and 26 scalars has modular weight -10 and unit plumbing residue. It is uniquely

$$\Phi(\Omega) = \frac{2\pi i}{\chi_{10}^{\text{note}}(\Omega)}. \quad (6.7)$$

To compare this with the implementation, define the genus-two theta constant

$$\theta \begin{bmatrix} a \\ b \end{bmatrix} (0|\Omega) = \sum_{\ell \in \mathbb{Z}^2} \exp[\pi i (\ell + a)^T \Omega (\ell + a) + 2\pi i (\ell + a)^T b] \quad (6.8)$$

and the raw product over the ten even characteristics,

$$\Psi_{10}(\Omega) = \prod_{\delta \text{ even}} \theta[\delta](0|\Omega)^2, \quad \chi_{10}^{\text{note}} = 2^{-12} \Psi_{10}. \quad (6.9)$$

Let $\epsilon = \Omega_{12}$. With the literal plumbing coordinate,

$$q = 2\pi i\epsilon + O(\epsilon^3), \quad \Psi_{10} = 2^{12}q^2\eta(\Omega_{11})^{24}\eta(\Omega_{22})^{24} + O(q^4). \quad (6.10)$$

Consequently $\chi_{10}^{\text{note}} \sim q^2\eta_1^{24}\eta_2^{24}$ and $d\epsilon = dq/(2\pi i)$. Substitution into (6.7) gives the unit chiral residue

$$\Phi d\Omega_{11}d\Omega_{22}d\epsilon \longrightarrow \frac{dq}{q^2} \frac{d\Omega_{11}}{\eta(\Omega_{11})^{24}} \frac{d\Omega_{22}}{\eta(\Omega_{22})^{24}}. \quad (6.11)$$

If one instead evaluates $(2\pi i)/\Psi_{10}$, its chiral residue is 2^{-12} . Its absolute square must therefore be multiplied by

$$\boxed{(2^{12})^2 = 2^{24}}. \quad (6.12)$$

This factor belongs to the local Mumford density; it is unrelated to either the sphere constant or the moduli-stack quotient.

6.4 The compact scalar at genus two

Normalize the holomorphic one-forms by $\oint_{A_I} \omega_J = \delta_{IJ}$ and $\oint_{B_I} \omega_J = \Omega_{IJ}$. A harmonic map to the circle is labelled by $m, n \in \mathbb{Z}^2$. Using $i \int_{\Sigma} \omega_I \wedge \bar{\omega}_J = 2Y_{IJ}$ in the scalar action gives

$$S_{m,n}^{(2)} = \pi r^2 (m + \bar{\Omega}n)^T Y^{-1} (m + \Omega n), \quad r = \frac{R_{\text{phys}}}{\sqrt{\alpha'}}. \quad (6.13)$$

Thus the classical lattice sum is

$$\boxed{\Theta_r^{(2)}(\Omega) = \sum_{m,n \in \mathbb{Z}^2} \exp[-\pi r^2 (m + \bar{\Omega}n)^T Y^{-1} (m + \Omega n)]}. \quad (6.14)$$

The connected compact zero mode is the target length, not merely the radius:

$$\int_0^{2\pi R_{\text{phys}}} dx_0 = 2\pi R_{\text{phys}} = 2\pi\sqrt{\alpha'} r. \quad (6.15)$$

If Z_X^{Ξ} is one noncompact scalar partition per unit target length in Xi's convention, the compact scalar is therefore

$$Z_{X,r}^{\Xi} = (2\pi\sqrt{\alpha'} r) Z_X^{\Xi} \Theta_r^{(2)}. \quad (6.16)$$

6.5 Liouville sewing and the Xi-normalized same-frame quotient

In a three-tube pants decomposition a , the normalized Liouville vacuum partition is defined entirely by the input two- and three-point data:

$$Z_L^{(a)} = \int_0^\infty \prod_{e=1}^3 \frac{dP_e}{\pi} C(P_1, P_2, P_3)^2 \left| \mathcal{F}_{25}^{(a)}(\mathbf{P}; \mathbf{q}) \right|^2. \quad (6.17)$$

The Virasoro block includes all descendant Gram matrices and the powers of q_e dictated by the weights. Equation (6.17) shows explicitly that there is no extra Liouville normalization constant: each internal edge carries exactly the completeness measure dP/π and each pair of pants carries one DOZZ coefficient.

Vacuum shifts and the meaning of A_L . Let $\mathcal{B}_X^{(a)}$ and $\mathcal{B}_L^{(a)}$ denote the unshifted scalar and Liouville blocks in the same three-tube plumbing channel, and define

$$\mathcal{V}_a := \prod_{e=1}^3 |q_e|^{-1/12}. \quad (6.18)$$

The Xi-normalized partitions are

$$Z_X^{\Xi,(a)} = A_X \mathcal{V}_a \mathcal{B}_X^{(a)}, \quad Z_L^{\Xi,(a)} = A_L^{\text{sewn}} \mathcal{V}_a^{25} \mathcal{B}_L^{(a)}, \quad A_X = \frac{1}{4\pi^2 \alpha'}, \quad A_L^{\text{sewn}} = 1. \quad (6.19)$$

Here A_X is Xi's two-handle momentum-measure normalization, whereas $A_L^{\text{sewn}} = 1$ follows from using literally the primary, metric, inverse metric, DOZZ coefficient, and intrinsic momentum P of Appendix I. Hence

$$\frac{Z_L^{\Xi,(a)}}{(Z_X^{\Xi,(a)})^{25}} = (4\pi^2 \alpha')^{25} \frac{\mathcal{B}_L^{(a)}}{(\mathcal{B}_X^{(a)})^{25}}. \quad (6.20)$$

The factor $\mathcal{V}_a^{25}$ cancels exactly. Thus the $|q_e|^{-25/12}$ shifts in the Liouville block produce no further constant; the surviving $(4\pi^2 \alpha')^{25}$ is the scalar normalization and must be combined with the critical 26-scalar seed and compact zero mode.

What genus one does and does not determine. Xi's equation (4.92) writes the unpunctured torus trace per ordinary Liouville length as

$$\frac{Z_L^{\Xi,(1)}}{V_\phi} = \int_0^\infty \frac{dP_{\text{phys}}}{\pi} \frac{e^{-\pi \alpha' \tau_2 P_{\text{phys}}^2}}{|\eta(\tau)|^2} = \frac{1}{2\pi \sqrt{\alpha' \tau_2} |\eta(\tau)|^2}. \quad (6.21)$$

The intrinsic CFT momentum of Appendix I instead gives

$$\mathcal{Z}_L^{\text{CFT},(1)} = \int_0^\infty \frac{dP_{\text{CFT}}}{\pi} \frac{e^{-4\pi \tau_2 P_{\text{CFT}}^2}}{|\eta(\tau)|^2} = \frac{1}{4\pi \sqrt{\tau_2} |\eta(\tau)|^2}. \quad (6.22)$$

Matching only the Gaussian characters gives

$$P_{\text{phys}} = \frac{2P_{\text{CFT}}}{\sqrt{\alpha'}}, \quad \frac{Z_L^{\Xi,(1)}}{V_\phi} = \underbrace{\frac{2}{\sqrt{\alpha'}}}_{J_\phi^{(1)}} \mathcal{Z}_L^{\text{CFT},(1)}. \quad (6.23)$$

The factor $J_\phi^{(1)}$ converts the dimensionful asymptotic wave number and the regulator defining the divergent ordinary length V_ϕ . It is not A_L^{sewn} and must not be inserted into the genus-two DOZZ integral. The symbol P is overloaded in the note: the DOZZ arguments in (4.119), the weight $h = 1 + P^2$, and every momentum in (6.17) remain the dimensionless intrinsic P_{CFT} . Replacing them by $2P_{\text{CFT}}/\sqrt{\alpha'}$ would change the arguments and weights of the non-homogeneous DOZZ function. Therefore

$$\boxed{J_\phi^{(1)} = \frac{2}{\sqrt{\alpha'}}, \quad A_L^{\text{sewn}} = 1.} \quad (6.24)$$

For the analytic derivation, place $Z_L^{(a)}$ together with Xi's noncompact scalar $Z_{X,k}^{\Xi,(a)}$ in the same conformal frame. Their Weyl transformations have the form

$$Z_{X,k}^{\Xi,(a)} = W_a Z_{X,k}^{\Xi,B}, \quad Z_L^{(a)} = W_a^{25} Z_L^B, \quad (6.25)$$

so the combination

$$\frac{Z_L^{(a)}}{(Z_{X,k}^{\Xi,(a)})^{25}} = \frac{Z_L^B}{(Z_{X,k}^{\Xi,B})^{25}} \quad (6.26)$$

is frame independent. This is why a separately guessed Weyl determinant must not be inserted into the integrand. The repository convention will be introduced only after the Xi-normalized density has been derived.

6.6 Xi-normalized momentum measure and matter replacement

Xi's note writes the physical momentum as k and fixes the free-boson states by

$$\boxed{\langle k|k'\rangle = 2\pi\delta(k - k'), \quad \mathbf{1} = \int_{\mathbb{R}} \frac{dk}{2\pi} |k\rangle\langle k|.} \quad (6.27)$$

This is the convention stated explicitly in Appendix P around (P.20). Appendix E gives the general form $C_X 2\pi\delta(k - k')$; Xi's convention here sets $C_X = 1$. All formulas through (6.37) use this convention.

The k_I are the two handle momenta, not the connected constant target-space mode x_0 . The latter is divided out for a noncompact scalar and becomes the target length $2\pi R_{\text{phys}} = 2\pi\sqrt{\alpha'} r$ for a compact scalar. Write

$$Y = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad D := \det Y = ac - b^2 > 0. \quad (6.28)$$

Xi's handle-momentum integral for one scalar per ordinary target-space volume is therefore

$$\begin{aligned} I_{0,k}^{\Xi}(Y) &:= \int_{\mathbb{R}^2} \frac{dk_1}{2\pi} \frac{dk_2}{2\pi} \exp[-\pi\alpha'(ak_1^2 + 2bk_1k_2 + ck_2^2)] \\ &= \frac{1}{(2\pi)^2} \int dk_1 dk_2 \exp\left[-\pi\alpha'a\left(k_1 + \frac{b}{a}k_2\right)^2 - \pi\alpha'\frac{D}{a}k_2^2\right] \\ &= \boxed{\frac{1}{4\pi^2\alpha'\sqrt{\det Y}}}. \end{aligned} \quad (6.29)$$

For bookkeeping inside the same Xi normalization, change variables only: $p_I = \sqrt{\alpha'} k_I$. Define

$$I_{0,p}^{(2\pi)}(Y) := \int_{\mathbb{R}^2} \frac{dp_1}{2\pi} \frac{dp_2}{2\pi} e^{-\pi p^T Y p} = \frac{1}{4\pi^2\sqrt{\det Y}}. \quad (6.30)$$

The genuine genus-two change of variables is then

$$\prod_{I=1}^2 \frac{dk_I}{2\pi} = \frac{1}{\alpha'} \prod_{I=1}^2 \frac{dp_I}{2\pi}, \quad \boxed{Z_{X,k}^{\Xi} = \frac{1}{\alpha'} Z_{X,p}^{(2\pi)}}. \quad (6.31)$$

No extra 2π occurs in this $k \rightarrow p$ map: Xi's factor $1/(2\pi)$ is present on both sides. If actual dimensionless- p states are introduced, they must be rescaled as $|p\rangle = \alpha'^{-1/4}|k = p/\sqrt{\alpha'}\rangle$ in order to retain $\langle p|p'\rangle = 2\pi\delta(p - p')$.

Now perform the matter replacement without referring to the repository:

$$Z_X^{26} \longrightarrow Z_{X,r}^{\Xi} Z_L, \quad Z_{X,r}^{\Xi} = (2\pi\sqrt{\alpha'} r) Z_{X,k}^{\Xi} \Theta_r^{(2)}, \quad (6.32)$$

and hence

$$\frac{Z_{X,r}^{\Xi} Z_L}{(Z_{X,k}^{\Xi})^{26}} = (2\pi\sqrt{\alpha'} r) \Theta_r^{(2)} \frac{Z_L}{(Z_{X,k}^{\Xi})^{25}}. \quad (6.33)$$

Using the raw theta product and temporarily omitting the external topology coefficient, the Xi-normalized local density is

$$\begin{aligned} \mathcal{D}_{2,\text{raw}}^{c=1,\Xi} &= \underbrace{\frac{|2\pi i|^2}{(4\pi^2 \alpha')^{26} (\det Y)^{13} |\Psi_{10}|^2}}_{\text{critical density in Xi's } dk/(2\pi) \text{ convention}} \underbrace{(2\pi\sqrt{\alpha'} r) \Theta_r^{(2)}}_{\text{compact zero mode and windings}} \underbrace{\frac{Z_L}{(Z_{X,k}^{\Xi})^{25}}}_{\text{Xi-normalized same-frame quotient}} \\ &= \boxed{\frac{2\pi r}{\sqrt{\alpha'}} (4\pi^2)^{-25} \frac{\Theta_r^{(2)}}{(\det Y)^{13} |\Psi_{10}|^2} \frac{Z_L}{(Z_{X,p}^{(2\pi)})^{25}}}. \end{aligned} \quad (6.34)$$

The second line merely uses the Xi-preserving change of variables (6.31):

$$|2\pi i|^2 (4\pi^2 \alpha')^{-26} \alpha'^{25} (2\pi\sqrt{\alpha'} r) = \frac{2\pi r}{\sqrt{\alpha'}} (4\pi^2)^{-25}. \quad (6.35)$$

Equivalently, the 26 critical scalars and the inverse 25-scalar quotient leave one genuine Xi $k \rightarrow p$ Jacobian,

$$(\alpha'^{-1})^{26} (\alpha'^{-1})^{-25} = \alpha'^{-1}. \quad (6.36)$$

Applying the unit-residue conversion (6.12) now defines the integrand before any code convention is mentioned:

$$\begin{aligned} \mathcal{I}_2^{\Xi}(\Omega; r, \alpha') &:= \boxed{2^{24} \frac{|2\pi i|^2}{(4\pi^2 \alpha')^{26}} \frac{(2\pi\sqrt{\alpha'} r) \Theta_r^{(2)}(\Omega)}{(\det Y)^{13} |\Psi_{10}(\Omega)|^2} \frac{Z_L(\Omega)}{(Z_{X,k}^{\Xi}(\Omega))^{25}}} \\ &= 2^{24} \frac{2\pi r}{\sqrt{\alpha'}} (4\pi^2)^{-25} \frac{\Theta_r^{(2)}(\Omega)}{(\det Y)^{13} |\Psi_{10}(\Omega)|^2} \frac{Z_L(\Omega)}{(Z_{X,p}^{(2\pi)}(\Omega))^{25}}. \end{aligned} \quad (6.37)$$

Translation to the repository evaluator. Only now introduce the repository's independent bare-measure object:

$$I_{0,p}^{\text{code}}(Y) := \int_{\mathbb{R}^2} dp_1 dp_2 e^{-\pi p^T Y p} = \frac{1}{\sqrt{\det Y}} = (2\pi)^2 I_{0,p}^{(2\pi)}(Y), \quad (6.38)$$

$$Z_{X,p}^{\text{code}} = (2\pi)^2 Z_{X,p}^{(2\pi)}. \quad (6.39)$$

The oscillator determinant is common to both objects. Combining this repository-only conversion with the Xi $k \rightarrow p$ Jacobian gives

$$\boxed{Z_{X,k}^{\Xi} = \frac{1}{\alpha'} Z_{X,p}^{(2\pi)} = \frac{1}{4\pi^2 \alpha'} Z_{X,p}^{\text{code}}}. \quad (6.40)$$

For clarity, the two factors are

$$C'_\alpha := \alpha'^{-1}, \quad C_{2\pi} := (2\pi)^{-2}, \quad C_{\text{code} \rightarrow k} := C'_\alpha C_{2\pi} = \frac{1}{4\pi^2 \alpha'}. \quad (6.41)$$

Here C'_α belongs to the Xi derivation; $C_{2\pi}$ belongs only to the final translation of the repository object. Substitution into (6.37) gives the numerically convenient but logically secondary form

$$\boxed{\mathcal{I}_2^{\Xi} = 2^{24} \frac{2\pi r}{\sqrt{\alpha'}} \frac{\Theta_r^{(2)}}{(\det Y)^{13} |\Psi_{10}|^2} \frac{Z_L}{(Z_{X,p}^{\text{code}})^{25}}}. \quad (6.42)$$

6.7 Gauge phase, sphere topology, and the final real coefficient

The critical six-form contains four independent factors: the phase $N_{2,0} = -i$, the Wick-rotation sign -1 , the form Jacobian $-8i$, and the coefficient $g_s^2 \alpha' / (8\pi)$. Their product is

$$(-i)(-1)(-8i) \frac{g_s^2 \alpha'}{8\pi} = \frac{g_s^2 \alpha'}{\pi}. \quad (6.43)$$

After stripping g_s^2 , the critical topology coefficient is α' / π . Replacing its sphere metric by the reduced physical $c = 1$ metric multiplies it by (4.5), hence

$$\underbrace{\frac{\alpha'}{\pi}}_{\text{critical real form}} \underbrace{\frac{2}{\alpha'}}_{\text{sphere sewing}} = \boxed{\frac{2}{\pi}}. \quad (6.44)$$

For comparison with the historical dimensionless-plumbing implementation, let $\mathcal{K}_2^{\text{old}}$ include its original critical coefficient α' / π , its dimensionless compact prefactor r , and the raw same-frame quotient $\mathcal{B}_L / (\mathcal{B}_X)^{25}$. The correlated Xi scalar conversion and the separate topology replacement are

$$\mathcal{N}_{\text{scalar}} = \frac{1}{2\pi\sqrt{\alpha'}}, \quad \mathcal{N}_{\text{top}} = \frac{2}{\alpha'}, \quad \boxed{\mathcal{N}_{\text{final}} = \mathcal{N}_{\text{scalar}} \mathcal{N}_{\text{top}} = \frac{1}{\pi \alpha'^{3/2}}}. \quad (6.45)$$

Thus $\mathcal{K}_2^{c=1} = \mathcal{N}_{\text{final}} \mathcal{K}_2^{\text{old}}$ in this explicitly defined comparison, and $\mathcal{N}_{\text{final}} = 1/\pi$ at $\alpha' = 1$. This relative multiplier must not be confused with the local coefficient $2/\pi$ below: the latter multiplies the already Xi-normalized object $\mathcal{I}_2^{\Xi}$, whereas $\mathcal{N}_{\text{final}}$ compares directly to the pre-conversion historical dimensionless-plumbing expression. The final positive-real local kernel is therefore

$$\boxed{K_2^{c=1}(\Omega; r, \alpha') = \frac{2}{\pi} \mathcal{I}_2^{\Xi}(\Omega; r, \alpha')}. \quad (6.46)$$

Equivalently, after the separate repository translation (6.40), the evaluator may display every factor as

$$\boxed{K_2^{c=1} = \underbrace{\frac{2}{\pi}}_{\text{topology/form}} \underbrace{2^{24}}_{\text{raw } \Psi_{10}} \underbrace{\frac{2\pi r}{\sqrt{\alpha'}}}_{\text{scalar measures}} \underbrace{\Theta_r^{(2)}}_{\text{compact lattice 26 Gaussians}} \underbrace{(\det Y)^{-13} |\Psi_{10}|^{-2}}_{\text{oscillators}} \underbrace{Z_L(Z_{X,p}^{\text{code}})^{-25}}_{\text{CFT replacement}}}. \quad (6.47)$$

When a coarse $PSp(4, \mathbb{Z})$ fundamental domain is used, a generic genus-two curve has the hyperelliptic involution. Dividing by this stabilizer gives a separate global weight $1/2$:

$$\boxed{\frac{\mathcal{A}_2(r)}{g_s^2} = \frac{1}{2} \int_{\mathcal{F}_2^{\text{coarse}}} d^3 X d^3 Y K_2^{c=1} = \frac{1}{\pi} \int_{\mathcal{F}_2^{\text{coarse}}} d^3 X d^3 Y \mathcal{I}_2^{\Xi}}. \quad (6.48)$$

The local $2/\pi$ and the global $1/2$ must each be applied exactly once.

6.8 Numerical reference at the self-dual radius

For reference, set $\alpha' = 1$ and $R_{\text{phys}} = r = 1$. The latest completed physical-mixture RQMC run used eight independent scrambled-Sobol replicates, four proposal components, and 256 proposals

per component; 6846 in-domain nodes required CFT evaluation. The archived table used the legacy dimensionless-scalar density $\mathcal{I}_2^{\text{old}}$ and the label

$$K_{2,\text{legacy}}^{\text{note}} = \frac{1}{\pi} \mathcal{I}_2^{\text{old}}, \quad \mathcal{I}_2^{\Xi} = \frac{1}{2\pi} \mathcal{I}_2^{\text{old}} = \frac{1}{2} K_{2,\text{legacy}}^{\text{note}} \quad (6.49)$$

and found

$$\int_{\mathcal{F}_2^{\text{coarse}}} d^3X d^3Y K_{2,\text{legacy}}^{\text{note}}(\Omega; 1) = (3.4977126795939466 \pm 0.08214314738188904) \times 10^{-4}. \quad (6.50)$$

After the exact scalar-measure conversion, the direct numerical reference for the Xi-normalized integrand defined in (6.37) is

$$\boxed{\int_{\mathcal{F}_2^{\text{coarse}}} d^3X d^3Y \mathcal{I}_2^{\Xi}(\Omega; 1) = (1.7488563397969733 \pm 0.04107157369094452) \times 10^{-4}.} \quad (6.51)$$

The uncertainty is the standard error across the eight complete replicates. Equation (6.51) is a coarse-domain integral and does not include the generic stack weight $1/2$. Combining it with the final local coefficient $2/\pi$ and that stack weight gives

$$\left. \frac{\mathcal{A}_2}{g_s^2} \right|_{r=1} = \frac{1}{\pi} \int_{\mathcal{F}_2^{\text{coarse}}} d^3X d^3Y \mathcal{I}_2^{\Xi} = (0.5566782624725754 \pm 0.013073487946953723) \times 10^{-4}. \quad (6.52)$$

This is the release field labelled `connected_logZ_genus2_over_gs_squared`. The corresponding coarse integral of $K_2^{c=1}$ is twice this value. No matrix-model normalization is used in any of these numbers. The machine-readable source is `output/data/genus2_c1_free_energy_direct_39/summary.json`.

6.9 One-to-one map to the numerical evaluator

At $\alpha' = 1$ the public code radius is r . The returned quantities correspond to the analytic factors as follows:

Implementation quantity	Analytic factor
<code>compact_target_zero_mode</code>	$2\pi r$
<code>compact_winding_sum</code>	$\Theta_r^{(2)}$
<code>det_im.omega</code>	$\det Y$
<code>chi10 with product</code>	Ψ_{10}
<code>liouville_partition</code>	Z_L from (6.17)
<code>noncompact_scalar_partition</code>	$Z_{X,p}^{\text{code}}$ with bare d^2p ; converted to Xi's physical- k object by (6.40)
<code>factorization_normalization</code>	2^{24}
<code>historical critical multiplier</code>	α'/π before the $c = 1$ sphere replacement
<code>c1_genus2_topology_correction</code>	$2/\alpha'$
<code>string_note_kernel_multiplier</code>	final sphere-normalized product $2/\pi$
Monte Carlo stack weight	$1/2$

The v3 production evaluator applies the scalar-measure conversion and the sphere-normalized multiplier exactly once. When archived values are read, their convention identifier selects an exact migration before integration; the stack weight remains a separate Monte Carlo factor.

7 Factor ledger

Genus one

Factor	Value	Origin
X zero mode	$R_{\text{phys}}/\sqrt{\alpha'}$	(5.5); also Poisson resummation
X classical action	$\pi R_{\text{phys}}^2 m\tau - n ^2 / (\alpha' \tau_2)$	action and torus metric
X oscillators	$\tau_2^{-1/2} \eta ^{-2}$	scalar determinant
Liouville trace per V_ϕ	$(2\pi\sqrt{\alpha'\tau_2} \eta ^2)^{-1}$	dP/π half-line Gaussian
ghost oscillators	$ \eta ^4$	left-right bc determinant
modulus/CKV	$d^2\tau/\tau_2$	(5.10)
torus stack weight	$1/2$	generic $z \mapsto -z$ stabilizer
$N_{1,0}$ and topology	1	plumbing unitarity and $\Lambda_1 = 1$

Genus two

Factor	Value	Origin
$N_{1,0}$	1	Plumbing unitarity, (3.3)
$N_{2,0}$	$-i$	Plumbing unitarity, (3.3)
$K_{S^2}^{\text{crit}}$	$8\pi/\alpha'$	BRST state metric and pole residue
$\tilde{K}_{S^2}$	$2/\sqrt{\alpha'}$	Physical $c = 1$ state unitarity
$\hat{K}_{S^2}^{c=1}$	4π	Fourier factor and (3.11)
Λ_2	$2/\alpha'$	Sphere topology sewing
one-scalar Gaussian	$(4\pi^2\alpha'\sqrt{\det Y})^{-1}$	(6.5)
26-scalar Gaussian	$(4\pi^2\alpha')^{-26}(\det Y)^{-13}$	product of 26 Gaussians
Mumford numerator	$ 2\pi i ^2 = 4\pi^2$	unit q -residue in Xi's convention
raw Ψ_{10} conversion	2^{24}	(6.12)
compact zero mode	$2\pi\sqrt{\alpha'} r$	target-circle length
compact lattice	$\Theta_r^{(2)}$	harmonic-map action
Xi state completeness	$dk/(2\pi)$	(6.27)
Xi $k \rightarrow p$ Jacobian	α'^{-1}	(6.31)
bare code $p \rightarrow (2\pi)$ -measure p	$(2\pi)^{-2}$	(6.39)
combined code $p \rightarrow \text{Xi } k$	$(4\pi^2\alpha')^{-1}$	(6.40)
code-form scalar product	$2\pi r/\sqrt{\alpha'}$	(6.42)
complex six-form Jacobian	$-8i$	(6.3)
final local coefficient	$2/\pi$	(6.44)
genus-two stack weight	$1/2$	generic hyperelliptic involution

8 Matrix model only as an external test

Nothing above uses the matrix model. After the worldsheet normalization has been fixed, the tree amplitude (3.13) may be compared with the collective-field result

$$\mathcal{A}_{1 \rightarrow 2}^{\text{MQM}} = \frac{i}{\mu} \omega \omega_1 \omega_2. \quad (8.1)$$

This gives the dictionary

$$\boxed{\mu^{-1} = 4\pi g_s^\Xi.} \quad (8.2)$$

If

$$f_2(r) = \frac{7r^2 + 10 + 7r^{-2}}{5760r}, \quad (8.3)$$

then the matrix-model prediction in the same coupling convention is

$$\frac{\mathcal{A}_2^{\text{MM}}(r)}{(g_s^\Xi)^2} = 16\pi^2 f_2(r). \quad (8.4)$$

Equation (8.4) is a test of (6.48), not an input used to alter it.

9 Items that still require an independent review

The derivation reduces the absolute normalization question to the following sharp checks.

1. Verify directly from Xi’s BPZ and physical-state conventions that the conversion from (3.10) to (3.12) contains exactly one Fourier 2π and exactly one Jacobian $\sqrt{\alpha'}$.
2. Verify that the scalar conversion (6.35) is included once and is not mistaken for the topology replacement (4.5) merely because both contain related zero-mode constants.
3. Verify each global $1/2$ against the precise moduli stack used in the numerical integration.
4. Test the analytic separating formula (6.10) numerically with the literal coordinate $uv = q$ and $q \simeq 2\pi i \Omega_{12}$.

If these checks pass, no additional moduli-independent factor is allowed by unitarity or by a change of local sewing coordinates.

10 Implementation manifest for the genus-two integral

The exhaustive, reviewer-facing source inventory is kept in `plumbing/genus2_monte_carlo_code_manifest.md`. The numerical path is not a single script. It has the following auditable stages:

$$\begin{aligned} \text{physical-measure RQMC nodes} &\longrightarrow \text{certified } (\Omega, q, \text{Sp}(4, \mathbb{Z})) \text{ chart} \\ &\longrightarrow (Z_L, Z_X, \Theta_r, \Psi_{10}) \longrightarrow K_2^{c=1} \longrightarrow \text{complete-scramble assembly.} \end{aligned} \quad (10.1)$$

The source closure of the pointwise evaluator contains 31 local Python modules. The manifest lists every one, together with the period-table builders, scheduler wrappers, radius reweighting and release code, and the independent convention checks. It also distinguishes code that changes the value from optional plots, recovery jobs, and historical samplers.

For the preferred physical-period-measure design the implemented estimator is

$$\left(\frac{\mathcal{A}_2}{g_s^2} \right)_s = \frac{1}{2} \frac{1}{4M} \sum_{a=1}^{4M} \mathbf{1}_{\mathcal{F}_2}(\Omega_{s,a}) \frac{J_Y(t_{1,s,a}, t_{3,s,a})}{p_{\text{mix}}(t_{1,s,a}, t_{3,s,a})} K_2^{c=1}(\Omega_{s,a}; r). \quad (10.2)$$

Thus no $\det(Y)^3$ multiplies the local kernel in the current production design. Out-of-domain proposals contribute exact zero; every in-domain node must have a certified CFT result; and the uncertainty is the standard error across independent complete scrambles. The exact control observable $K_{\text{control}} = \det(Y)^{-3}$ must reproduce

$$\int_{\mathcal{F}_2^{\text{coarse}}} \frac{d^3 X d^3 Y}{\det(Y)^3} = \frac{\pi^3}{270}. \quad (10.3)$$

These requirements are enforced by the design generator and strict assembler, not added by post-processing.

References

- [1] X. Yin, *String Theory Notes*, especially Sections 3.8, 4.6, 4.8–4.10 and Appendices E, I.5, and P; equations (4.32), (4.58)–(4.72), (4.90)–(4.122), (I.39)–(I.42), and (P.20).
- [2] B. Balthazar, V. A. Rodriguez, and X. Yin, *The $c = 1$ String Theory S-Matrix Revisited*, arXiv:1705.07151.
- [3] E. D'Hoker and D. H. Phong, *Two-Loop Superstrings VI: Non-Renormalization Theorems and the 4-Point Function*, and related genus-two measure papers.